

TRAFFIC LIGHT COUNT-DOWN TIMER WITH DUAL-COLOUR DISPLAY

■ VINAY CHADDHA

Normally, count-down timers of traffic lights display the count-down time only in red colour. The drawback here is that the drivers may get confused when the signal light is green but the timer display is in red.

Here we describe a dual-colour traffic count-down timer that displays count-down time in either red or green

colour matching with the signal light. It is a 3-digit timer based on AVR Atmega328 microcontroller. The timer board can retrofit into existing signal lights.

Other key features of this timer are constant display intensity despite voltage variation, low power consumption (which makes it suitable for battery-powered systems), and an optional feature for intensity variation with time of the day (high intensity



in daytime, medium intensity during dusk and dawn, and low intensity at night).

How does count-down timer works?

All retrofit timers used in traffic light systems work by learning the traffic light 'on' and 'off' time first. Learning

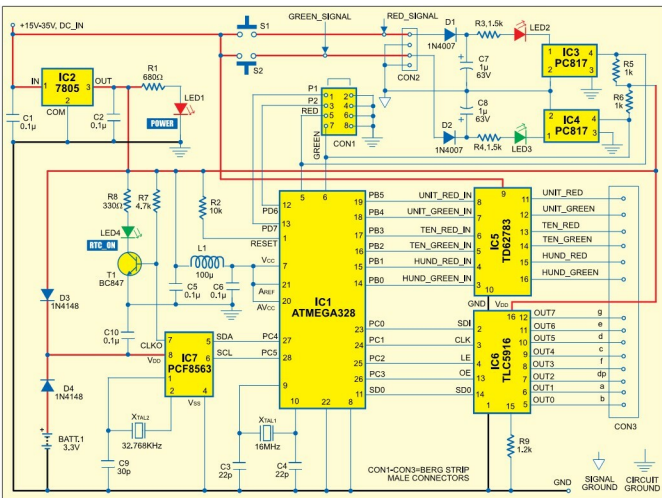


Fig. 1: Circuit of AVR-based traffic light count-down timer